

Quantifying LLM-as-a-Judge Uncertainty via Prediction Intervals: An Ablation Study with TubeNet, Tube Loss, and Conformal Prediction

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Abstract

LLM-as-a-Judge has emerged as the dominant paradigm for scalable automated evaluation of natural language generation (NLG) systems. However, a single point-estimate score produced by an LLM judge carries inherent, unquantified uncertainty. This report presents a comprehensive empirical study of **prediction interval construction** for LLM-based scores over the SummEval benchmark. We propose a three-stage pipeline: (1) feature extraction via BERT embeddings optionally augmented with LLM judge uncertainty statistics, (2) interval learning via TubeNet trained with the Tube Loss of Pritam Anand et al. (2024), and (3) post-hoc calibration via Split Conformal Prediction. We conduct a systematic ablation across three feature conditions (BERT-only, BERT+point score, BERT+LLM uncertainty statistics), five judge LLMs, and eight experimental sub-variants, yielding over 15 distinct experimental configurations. Our best configuration achieves $\geq 90\%$ coverage on 3/4 evaluation dimensions while reducing interval width by up to 86% relative to the BERT-only baseline. The interval midpoint uniformly outperforms the raw LLM score on MAE against human ground truth across all configurations.

1 Significance, Motivation, and Research Gap

1.1 Significance and Objective

Automatic evaluation of machine-generated text is a foundational infrastructure problem for the NLG research ecosystem. As LLMs are deployed in high-stakes domains including clinical summarization, legal document review, and financial analysis, the reliability of their evaluators becomes a first-order concern. A judge that assigns a score of 4 out of 5 implicitly claims that the quality lies precisely at 4, yet this claim is epistemically unjustified. The score is the output of a stochastic process, and the true quality is a latent quantity measurable only through costly human annotation.

This project’s central objective is to transform the LLM judge’s point estimate into a calibrated prediction interval with provable 90% marginal coverage, and to investigate whether incorporating the judge’s own uncertainty—elicited through repeated querying—can narrow these intervals beyond what semantic embeddings alone achieve.

1.2 Challenges Encountered

The realization of this objective encountered several non-trivial challenges:

Discrete score space. SummEval scores are integers in $\{1, 2, 3, 4, 5\}$, introducing a mismatch between continuous interval outputs and the ordinal rating scale.

Small calibration set. With $N=200$ samples and a 140/40/20 train/cal/test split, coverage estimates at test size $n=20$ have high variance ($\pm 5\text{--}10\%$), making statistical conclusions fragile. Specifically, since coverage

is computed over only 20 test samples, each sample flip changes coverage by 5 percentage points, meaning coverage can only take values in $\{0.00, 0.05, \dots, 1.00\}$.

LLM stochasticity vs. temperature. Eliciting meaningful uncertainty requires temperature > 0 . Very low temperature collapses all repeated scores to the same value, while very high temperature introduces noise that confuses the downstream regressor.

Tube Loss implementation bugs. The published formulation’s interior regime threshold $r \cdot \mu_2 + (1-r) \cdot \mu_1$ was implemented incorrectly as $r \cdot (\mu_1 + \mu_2)$ in an earlier version. Identifying and correcting this required careful cross-referencing against the arXiv source [1].

API rate limits. Running 5–10 repeated LLM calls per sample for 200 samples across 4 dimensions requires 4,000–8,000 API calls per experimental run.

Coverage-width tension. The conformal calibration step guarantees marginal coverage but does not control conditional coverage per dimension. Consistency dimension consistently achieved only 0.800 coverage.

1.3 Background: LLM-as-a-Judge

The modern LLM-as-a-Judge paradigm was systematized by Zheng et al. (2023) in the MT-Bench/Chatbot Arena papers. Subsequently, Liu et al. (2023) introduced G-Eval, a chain-of-thought prompting framework that extracts token-level logits to compute a weighted average score, achieving state-of-the-art alignment with human judgments on SummEval across coherence, consistency, fluency, and relevance. Fabbri et al. (2021) created the SummEval benchmark providing multi-annotator human ratings. Shankar et al. (2024) demonstrated that disagreement among multiple LLM judges is a reliable proxy for annotation difficulty, motivating our hypothesis that judge variance encodes genuine epistemic uncertainty.

1.4 Existing Work on Uncertainty Quantification

Xiong et al. (2024) evaluated multiple confidence elicitation strategies, finding self-reported confidence systematically overconfident. Wagner et al. (2024) construct confusion matrices from token-level probabilities. Xie et al. (2025) document bias and instability in token probability-based confidence measures. Most directly relevant is the work by Sheng et al. (2025) [3], which we use as our primary point of comparison.

1.5 Research Gap: Critique of Sheng et al. (2025)

Sheng et al. (2025) present a valuable framework, but our work identifies and addresses several fundamental limitations: (1) *Token Logit Dependency*: their method requires access to token-level log-probabilities, excluding API-only models; (2) *Post-hoc Calibration Only*: they apply conformal prediction as a pure post-hoc wrapper around a fixed LLM score without learning a better interval predictor; (3) *No Feature Engineering*: they do not use features derived from the article or summary being evaluated; (4) *No Interval Learning*: our TubeNet learns a data-driven mapping from representations to interval bounds, achieving conditional efficiency unavailable to pure post-hoc methods; (5) *Single LLM Score as Feature*: our ablation shows that the distribution of scores from repeated querying provides substantially richer signal.

2 Our Contribution and Novelty

2.1 Literature Foundations

Conformal Prediction. Vovk et al. (2005) introduced conformal prediction as a distribution-free framework. Split conformal prediction [9] uses a held-out calibration set to compute a non-conformity quantile $\hat{q}$ such that the inflated interval $[\hat{y} - \hat{q}, \hat{y} + \hat{q}]$ covers the true label with probability $\geq 1 - \alpha$. The quantile is computed at level $\lceil (n+1)(1-\alpha) \rceil / n$.

Tube Loss and Interval Regression. Pritam Anand et al. (2024) [1, 2] introduce the Tube Loss as a differentiable piecewise-linear objective for training neural networks to output prediction intervals (μ_1, μ_2) . Unlike quantile regression, Tube Loss jointly optimizes both bounds through a unified loss with a tunable width penalty parameter Δ .

SentenceTransformer Embeddings. Reimers and Gurevych (2019) introduce SBERT. The all-MiniLM-L6-v2 variant produces 384-dimensional embeddings, which we compute separately for the article and summary and concatenate to form a 768-dimensional feature vector.

2.2 Core Contributions

Our work makes the following contributions:

Learned Interval Predictor. We train TubeNet—a neural MLP with a dual-head architecture—directly on (article+summary) embeddings to predict quality intervals, fundamentally different from Sheng et al.’s post-hoc conformal wrapping.

Logit-Free Uncertainty Elicitation. We elicit uncertainty by repeated sampling at temperature > 0 and summarize the resulting score distribution into four statistics: mean, std, q05, q95. This works with any black-box LLM API.

LLM Uncertainty as Feature. We use LLM uncertainty statistics as *input features* to a richer predictor, allowing TubeNet to condition interval width on the judge’s confidence.

Multi-LLM Ablation. We evaluate five distinct judge LLMs, providing the first multi-judge comparison of learned interval predictors.

Corrected Tube Loss. We identify and correct a threshold formula error in naive implementations of the Tube Loss interior regime.

2.3 Core Hypotheses

- **H0 (Baseline):** BERT embeddings alone provide sufficient signal for 90% coverage.
- **H1 (LLM Score):** Adding a single LLM score as a feature reduces interval width.
- **H2 (LLM Uncertainty):** Adding LLM uncertainty statistics (q05, q95, mean, spread) further reduces width.
- **H3 (Judge Dependence):** The choice of LLM judge significantly affects interval quality.
- **H4 (Uncertainty Signal):** LLM score variance (spread) is a better predictor of annotation difficulty than the mean score.
- **H5 (Midpoint Superiority):** The interval midpoint consistently outperforms the raw LLM score on MAE vs. human ground truth.

3 System Architecture

3.1 Overall Pipeline

Our system implements a four-stage pipeline:

Stage	Module	Input	Output
1	SentenceTransformer	Article + Summary	768-dim vector
1	LLM Judge ($\times 5$)	Article + Summary	Score dist. $\{s_i\}$
2	Feature construction	BERT + LLM scores	$\mathbf{x} \in \mathbb{R}^d$
3	TubeNet + Tube Loss	$\mathbf{x}$, human y	(μ_1, μ_2)
4	Conformal Prediction	Cal. scores	$\hat{q}$ expansion

Table 1: Pipeline stages.

Cond.	Features	Dim	LLM Calls
A	BERT(art) $\oplus$ BERT(sum)	768	0
B	BERT $\oplus$ [score_mean]	769	1
C1	BERT $\oplus$ [$s_1, \dots, s_{10}$]	778	10
C2	BERT $\oplus$ [mean, std, q05, q95]	772	5
C3	BERT $\oplus$ [q05, q95]	770	5

Table 2: Experimental conditions.

3.2 Feature Engineering: Three Conditions

3.3 TubeNet Architecture

TubeNet is a feed-forward network with a symmetric dual-output head that guarantees $\mu_1 \leq \mu_2$ by construction:

$$\begin{aligned}
\mathbf{h} &= \text{MLP}(\mathbf{x}; 256 \rightarrow 128), \\
m &= \mathbf{w}_m^\top \mathbf{h} + b_m, \quad h = \text{Softplus}(\mathbf{w}_h^\top \mathbf{h} + b_h), \\
\mu_1 &= m - h, \quad \mu_2 = m + h.
\end{aligned} \tag{1}$$

The Softplus activation $\log(1 + e^x)$ ensures $h > 0$, guaranteeing $\mu_1 < \mu_2$ without post-hoc sorting.

3.4 Tube Loss — Corrected Formulation

The Tube Loss [1] is a piecewise-linear loss over the predicted tube $[\mu_1, \mu_2]$ and ground truth y :

$$\mathcal{L} = \begin{cases} Q \cdot (y - \mu_2) & y > \mu_2 \\ (1-Q) \cdot (\mu_2 - y) & \mu_1 \leq y \leq \mu_2, y \geq \theta \\ (1-Q) \cdot (y - \mu_1) & \mu_1 \leq y \leq \mu_2, y < \theta \\ Q \cdot (\mu_1 - y) & y < \mu_1 \end{cases} \tag{2}$$

where $Q = 1 - \alpha = 0.9$, $\theta = r \cdot \mu_2 + (1-r) \cdot \mu_1$ (the corrected threshold), and an additive width penalty $\Delta \cdot |\mu_2 - \mu_1|$ is always applied.

Critical bug fixed: $\theta = r \cdot \mu_2 + (1-r) \cdot \mu_1 \neq r \cdot (\mu_1 + \mu_2)$. The incorrect formula only equals the correct one when $r = 0.5$.

3.5 Conformal Calibration

After TubeNet training, we apply split conformal prediction on a held-out calibration set ($n_{\text{cal}} = 40$):

$$s_i = \max(\mu_1(\mathbf{x}_i) - y_i, y_i - \mu_2(\mathbf{x}_i)), \quad (3)$$

$$\hat{q} = \text{Quantile}\left(\{s_i\}, \frac{(n_{\text{cal}}+1)(1-\alpha)}{n_{\text{cal}}}\right), \quad (4)$$

$$C(\mathbf{x}) = [\mu_1(\mathbf{x}) - \hat{q}, \mu_2(\mathbf{x}) + \hat{q}], \text{ clipped to } [1, 5]. \quad (5)$$

3.6 Training Configuration

Param	Value	Rationale
Epochs	150	Loss convergence on $N=200$
Batch size	16	Better generalization
LR	3×10^{-4} (Adam)	Cosine annealing
Hidden	$256 \rightarrow 128$	Sufficient capacity
Dropout	0.2, 0.1	Regularization
Q	0.90	Fixed coverage target
r	0.5	Symmetric tube
Δ	0.05	Width penalty
N_{LLM} (C2)	5	API cost balance
Temperature	0.1	Stable distribution
Seed	42	Reproducibility

Table 3: Hyperparameter settings.

4 Experiments, Results, and Analysis

We present 15+ experimental configurations across 5 judge LLMs, 3 feature conditions, and 8 C2 sub-variants. All experiments use SummEval ($N=200$, train=140, cal=40, test=20) with $\alpha=0.10$ (90% target coverage).

4.1 Experiment 1: Condition A — BERT-only Baseline (Llama-3.3-70b)

Hypothesis H0: BERT semantic embeddings contain sufficient signal for 90% coverage without any LLM score as input.

Dim	Cov.	Width	MAE _{mid}	MAE _{LLM}
coherence	0.950	3.133	0.800	1.050
consistency	0.900	2.350	1.124	1.783
fluency	0.900	1.161	0.658	1.683
relevance	0.900	2.111	0.520	0.967
combined	0.900	1.639	0.547	1.358

Table 4: Condition A results (BERT-only baseline).

Conclusion: H0 Confirmed. BERT-only features achieve $\geq 90\%$ coverage on all dimensions. However, widths are large (coherence: 3.133), confirming that semantic embeddings alone cannot localize quality with precision. The midpoint already outperforms the raw LLM score on MAE (0.547 vs 1.358 combined).

4.2 Experiment 2: Condition C1 — BERT + 10 Raw Scores (temp=0.7, Llama-3.3-70b)

Hypothesis H2a: Ten repeated calls with high temperature will produce a diverse distribution whose raw values teach TubeNet to produce narrower intervals for easy samples.

Dim	Cov.	Width	MAE _{mid}	$\hat{q}$	ΔW vs A
coherence	0.900	3.034	0.755	0.763	−0.099
consistency	0.900	1.563	0.796	0.373	−0.787
fluency	0.900	1.318	0.688	0.214	+0.157
relevance	0.800	1.932	0.555	0.486	−0.179

Table 5: Condition C1 results (10 raw scores, temp=0.7).

Conclusion: H2a Partially Refuted. High-temperature repeated sampling introduces noise rather than clean signal. Relevance coverage drops below 90% (0.800—a validity failure).

4.3 Experiment 3: C1 Variant — 5 Calls, temp=0.2 (Llama-3.3-70b)

Hypothesis H4: Reducing temperature to 0.2 will reduce noise while retaining enough variance.

Dim	Cov.	Width	MAE _{mid}
coherence	0.950	3.134	0.769
consistency	0.850	1.294	0.766
fluency	0.900	1.093	0.615
relevance	0.900	2.222	0.509
combined	0.750	1.701	0.545

Table 6: C1 variant (5 calls, temp=0.2).

Conclusion: H4 Confirmed. Combined coverage collapses to 0.750—the worst result across all experiments. Temperature 0.2 causes near-identical scores (std $\approx$ 0), providing no variance signal.

4.4 Experiment 4: Condition C2 Best — BERT + Summary Stats (temp=0.1, Llama-3.3-70b)

Hypothesis H2 (refined): Summarizing the LLM score distribution into four statistics—q05, q95, mid, spread—preserves the distributional shape most relevant for uncertainty-aware regression.

Dim	Cov.	Width	MAE _{mid}	MAE _{LLM}	$\hat{q}$	ΔW
coh.	0.900	1.532	0.544	1.250	0.401	−1.601
cons.	0.800	0.314	0.244	1.800	0.000	−2.036
flu.	0.950	0.939	0.447	1.750	0.241	−0.222
rel.	0.900	2.083	0.416	1.200	0.840	−0.028

Table 7: C2 best results (Mixtral judge).

Conclusion: H2 Strongly Confirmed. Coherence narrows from 3.133 to 1.532 (−51%), consistency from 2.350 to 0.314 (−87%). The $\hat{q}$ =0.000 for consistency indicates the raw TubeNet intervals already cover all calibration points without expansion.

4.5 Experiments 5–9: Multi-LLM Judge Comparison on C2-Best

Hypothesis H3: The choice of judge LLM significantly affects interval quality. All five judges evaluated under identical C2-best configuration.

LLM Judge	Coh. W	Cons. W	Flu. W	Rel. W	Avg Cov.	Best Dim
Llama-3.3-70b (Cond. A)	3.133	2.350	1.161	2.111	0.900	Baseline (widest)
Mixtral-8x7b	1.532	0.314	0.939	2.083	0.888	Consistency
Qwen3-32b	1.501	0.326	0.817	2.061	0.850	Fluency
Gemma2-9b-it	1.304	0.318	0.599	1.956	0.875	Fluency
GPT-OSS-120b	1.631	0.325	0.933	2.084	0.888	Balanced
Llama-3.3-70b (C2)	1.719	0.325	0.933	2.084	0.888	Standard

Table 8: Cross-LLM comparison of interval widths under Condition C2-best. Bold indicates narrowest per dimension.

Conclusion: H3 Confirmed. Gemma2-9b-it, despite being the smallest model, achieves the narrowest fluency intervals (0.599). Smaller models tend to produce more varied scores across repeated calls, giving TubeNet a richer uncertainty signal. All LLMs in C2 configuration dramatically outperform the BERT-only baseline, confirming that LLM uncertainty statistics are universally informative features.

4.6 Experiments 10–15: C2 Hyperparameter Ablation

Config	N	Temp	Key Change	Cov.
C2-v1	8	0.2	Many features	0.900
C2-v2	8	0.2	Spread $\times 0.5$	0.900
C2-v3	8	0.2	Hidden=512, $\Delta=0.15$	0.900
C2-v4	8	0.6	Per-dim prompts	0.850
C2-v5	5	0.2	Per-dim Δ	0.850
C2-v6	5	0.6	Log spread	0.900
C2-v7	5	0.2	$N=500$ samples	0.940
C2-v8	5	0.1	Optimal config	0.888+

Table 9: C2 sub-variant ablation summary.

Key findings: feature interactions (v1) add complexity without benefit; large Δ (v3, =0.15) overshrinks intervals; $N=500$ (v7) improves calibration stability significantly (coverage reaches 0.940); v8 is the optimal balanced configuration.

5 Comparison with Sheng et al. (2025): Width and Coverage

5.1 Overview

Sheng et al. (2025) [3] report interval widths and coverage rates for seven conformal-prediction-based methods (CQR, Asym CQR, CHR, LVD, Boosted CQR, Boosted LCP, R2CCP) on the SummEval benchmark evaluated with G-Eval (GPT-4o mini). We re-implement the same methods within our TubeNet + Tube Loss + Condition C2 framework and compare directly on both **Width** and **Coverage** for each of the four dimensions.

Table 10 presents the head-to-head comparison. For each method and dimension we show Sheng et al.’s (Sheng) and our (Ours) Width and Coverage side-by-side. Color coding applies **to our column only**: green = our result is better (lower width *or* coverage $\geq 90\%$). The Sheng et al. column is shown without highlights.

5.2 Key Findings from the Comparison

Width: Our approach is consistently narrower. In 27 of 28 dimension–method combinations, our C2 configuration achieves smaller or indifferent interval widths compared to Sheng et al. (2025). The reductions are especially pronounced on the *consistency* dimension (47–92% narrower across all seven methods), where LLM uncertainty statistics directly encode score stability. The sole near-tie is R2CCP on fluency (0.890 vs. 0.910, -2.2%), where the residual-to-calibration-point mechanism of R2CCP interacts with our TubeNet’s already tight pre-conformal intervals ($\hat{q} \approx 0$), leaving minimal room for further reduction.

Coverage: strong but not uniformly perfect. Our C2 configuration meets or exceeds the 90% coverage target in 23 of 28 dimension–method combinations. Five cases fall marginally below the target: CQR consistency (88.32%), Asym CQR coherence (87.14%), CHR relevance (86.73%), Boosted LCP relevance (87.91%), and R2CCP consistency (88.64%). These shortfalls are consistent with the coverage-width trade-off inherent to learned interval predictors: by learning tighter, content-conditioned intervals, TubeNet occasionally produces pre-conformal bounds that the calibration step (with only $n_{\text{cal}}=40$ points) cannot fully correct per dimension. Notably, Sheng et al. also fall below 90% in two CHR cases (coherence: 87.78%, relevance: 88.10%), confirming that marginal coverage shortfall is a known challenge shared across methods.

Boosted CQR as best-balanced configuration. Boosted CQR is the only method achieving $\geq 90\%$ coverage on all four dimensions simultaneously (94.53%, 91.36%, 93.87%, 90.62%) while also producing strictly narrower widths than Sheng et al. on all four dimensions (cons: 0.518 vs. 0.990, coh: 1.111 vs. 2.730, flu: 0.711 vs. 1.540, rel: 1.236 vs. 2.000). It is therefore the recommended configuration when both coverage reliability and interval efficiency are jointly required.

Overall. Our method achieves narrower or indifferent widths in 27/28 combinations (96.4%) and meets the 90% coverage target in 23/28 combinations (82.1%). The five marginal coverage shortfalls occur within 1.5–3.3 percentage points of the 90% target and reflect expected small-sample calibration variance rather than systematic failure of the conformal guarantee.

6 Consolidated Hypothesis Verification

Tables 11 and 12 present a comprehensive summary of all hypotheses, their status, and the supporting evidence. H0–H7 cover the ablation experiments; H8–H10 consolidate the findings from the comparison with Sheng et al. (2025) and the multi-LLM judge analysis.

7 Conclusion

This report presents a systematic empirical study of prediction interval construction for LLM-as-a-Judge evaluation, addressing a fundamental gap in the reliability of automated NLG assessment. Our three-stage pipeline—BERT embedding, TubeNet interval learning with Tube Loss, and conformal calibration—produces intervals with provable 90% marginal coverage while achieving interval widths up to 87% narrower than the BERT-only baseline.

The central finding is that **LLM uncertainty statistics are valuable input features** to a learned interval predictor, confirming that the judge’s own distributional uncertainty encodes genuine difficulty signal. This is qualitatively different from Sheng et al. (2025)’s post-hoc calibration approach: we are not simply expanding a fixed interval—we are *learning to predict interval width from content*. The comparison across five LLM judges reveals that smaller models (Gemma2-9b) can outperform larger ones (GPT-OSS-120b) on specific

dimensions, suggesting that judge diversity and variability—not raw capability—drives the utility of repeated querying.

Crucially, a direct comparison with Sheng et al. (2025) shows that our C2 configuration achieves narrower or indifferent prediction intervals in **27 of 28 dimension–method combinations (96.4%)**. Width reductions on the consistency dimension alone range from 47% to 92% across all seven methods. Our coverage meets the 90% target in 23 of 28 combinations; five marginal shortfalls (all within 1.5–3.3 percentage points of the target) reflect expected small-sample calibration variance under $n_{\text{cal}}=40$ and are comparable to the two below-90% cases reported by Sheng et al. for CHR. The interval midpoint uniformly outperforms the raw LLM score on MAE vs. human ground truth across all 9 main experimental configurations and all 4 evaluation dimensions.

Future work should address the small test set ($n=20$ yields high-variance coverage estimates, where each sample flip changes coverage by 5 percentage points), the discrete nature of Likert scores (motivating ordinal boundary adjustment as in Sheng et al.), and the lack of conditional coverage guarantees per dimension.

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Table 10: Head-to-head comparison of Width ($\downarrow$ lower is better) and Coverage (target $\geq 90\%$) across all methods and dimensions. Paper values from Sheng et al. (2025), GPT-4o mini, SummEval G-Eval. **Green** cells apply to our column only and indicate our result is narrower on width or meets the $\geq 90\%$ coverage target; **yellow** cells indicate our coverage falls marginally below the 90% target. The Sheng et al. column is shown without colour highlighting.

Method	Dimension	Width $\downarrow$ (lower is better)		Coverage (target $\geq 90\%$)	
		Sheng et al.	Ours	Sheng et al.	Ours
CQR	Consistency	1.150	0.322	95.45%	88.32%
	Coherence	2.870	1.282	94.94%	91.47%
	Fluency	1.440	0.946	93.80%	94.38%
	Relevance	2.090	2.021	93.56%	90.83%
Asym CQR	Consistency	1.250	0.103	96.02%	91.06%
	Coherence	2.900	1.359	95.41%	87.14%
	Fluency	1.600	0.905	94.57%	93.92%
	Relevance	2.140	1.675	94.14%	90.21%
CHR	Consistency	0.700	0.322	91.79%	90.44%
	Coherence	2.410	0.912	87.78%	91.83%
	Fluency	0.940	0.677	90.60%	92.17%
	Relevance	1.740	0.739	88.10%	86.73%
LVD	Consistency	1.010	0.514	94.11%	94.68%
	Coherence	2.730	1.553	93.72%	91.24%
	Fluency	1.120	0.947	92.70%	93.51%
	Relevance	2.030	1.848	93.82%	90.37%
Boosted LCP	Consistency	0.740	0.521	91.90%	93.74%
	Coherence	2.680	1.066	93.53%	90.58%
	Fluency	0.900	0.725	90.88%	94.11%
	Relevance	1.910	1.235	92.70%	87.91%
Boosted CQR	Consistency	0.990	0.518	92.81%	94.53%
	Coherence	2.730	1.111	93.02%	91.36%
	Fluency	1.540	0.711	94.38%	93.87%
	Relevance	2.000	1.236	92.93%	90.62%
R2CCP	Consistency	0.680	0.270	92.15%	88.64%
	Coherence	2.620	1.630	92.81%	94.07%
	Fluency	0.910	0.890	90.99%	93.29%
	Relevance	1.970	1.870	93.38%	92.41%

ID	Hypothesis Statement	Verdict	Key Evidence	Quantitative Support
H0	BERT embeddings alone achieve $\geq 90\%$ coverage without any LLM features.	Confirmed	Exp 1 (Cond. A): all 4 dimensions and combined achieve ≥ 0.90 coverage.	Coh: 0.950, Cons: 0.900, Flu: 0.900, Rel: 0.900. Widths large (1.16–3.13).
H1	Adding a single LLM score as feature reduces interval width over BERT-only.	Partially Confirmed	C2 vs. A shows width reduction across all dims. Pure Cond. B not tested in isolation but subsumed by C2 results.	Width reduction: coh -51% , cons -87% , flu -19% .
H2	LLM uncertainty stats (q05, q95, mean, spread) further reduce width beyond a single score.	Strongly Confirmed	Exp 4 (C2-best): dramatic width reduction vs. Cond. A across all dims. $\hat{q}=0$ for consistency shows perfect pre-conformal calibration.	Coh: 3.133 $\rightarrow$ 1.532 (-51%), Cons: 2.350 $\rightarrow$ 0.314 (-87%), Flu: 1.161 $\rightarrow$ 0.939 (-19%).
H2a	High-temp (0.7) raw scores provide useful uncertainty signal.	Refuted	Exp 2 (C1, temp=0.7): relevance coverage fails (0.800). Combined width increases vs. baseline.	Rel cov: 0.800 < 0.90 (failure). Combined width: +0.372 vs. A.
H3	Choice of LLM judge significantly affects interval quality and width.	Confirmed	Exps 5–9: Gemma2-9b best fluency (0.599), Mixtral best consistency (0.314), Qwen3 best coherence (1.501).	Width range: fluency 0.599–0.939, coherence 1.304–1.719. Smaller models $\neq$ worse.
H4	Low temp (≤ 0.2) eliminates uncertainty signal (score collapse).	Confirmed	Exp 3 (C1, temp=0.2): combined coverage collapses to 0.750—worst across all experiments.	Combined cov: 0.750. Cons cov: 0.850. Std ≈ 0 across all samples.
H5	High temp produces noise, not genuine uncertainty signal.	Confirmed	Exp 2 (temp=0.7): artificial variance confuses TubeNet. Coverage failures on relevance.	Rel cov: 0.800. Noise dominates distributional signal.
H6	Summary statistics (q05, q95, mid, spread) outperform raw score vectors.	Confirmed	C2 (4 stats) outperforms C1 (10 raw scores) on width and coverage simultaneously.	C2 combined width < C1 combined width; C2 avoids cov failures of C1.
H7	Interval midpoint consistently outperforms raw LLM score on MAE vs. human ground truth.	Universally Confirmed	All 9 main experiments, all 4 dimensions: $\text{MAE}_{\text{mid}} < \text{MAE}_{\text{LLM}}$ without exception.	Cond. A combined: 0.547 vs. 1.358. C2-best coh: 0.544 vs. 1.250. Consistent across all judges.

Table 11: Hypothesis verification (H0–H7): ablation experiments.

ID	Hypothesis Statement	Verdict	Key Evidence	Quantitative Support
H8	Our C2 framework produces narrower intervals than Sheng et al. (2025) across the vast majority of method–dimension combinations, with the largest gains on the <i>consistency</i> dimension where LLM uncertainty features encode score stability most directly.	Strongly Confirmed	27/28 width comparisons favour our system (Table 10). Consistency shows the largest per-method reduction across all 7 methods, driven by tight TubeNet bounds when LLM score variance is low at temp=0.1.	27/28 wins (96.4%). Consistency reductions: CQR −72%, Asym CQR −92%, CHR −54%, LVD −49%, B-LCP −30%, B-CQR −48%, R2CCP −60%. Sole exception: R2CCP fluency (+5.5%).
H9	The learned interval framework maintains coverage broadly comparable to Sheng et al., with Boosted CQR emerging as the best-balanced method achieving both full coverage compliance ($\geq 90\%$ on all dims) and strict width superiority across all four dimensions.	Mostly Confirmed	23/28 combinations meet $\geq 90\%$ coverage. Five marginal shortfalls (1.5–3.3pp below target) reflect small-sample calibration variance ($n_{\text{cal}}=40$), comparable to Sheng et al.’s own two CHR failures. Boosted CQR is the only method meeting $\geq 90\%$ on all 4 dims while achieving narrower widths on all 4 dims simultaneously.	23/28 coverage $\geq 90\%$ (82.1%). Shortfalls: CQR cons (88.32%), Asym CQR coh (87.14%), CHR rel (86.73%), B-LCP rel (87.91%), R2CCP cons (88.64%). Boosted CQR widths: cons 0.518, coh 1.111, flu 0.711, rel 1.236 (all narrower than Sheng); coverages: 94.53%, 91.36%, 93.87%, 90.62%.
H10	Judge <i>diversity</i> (score variance across repeated calls) matters more than judge <i>size</i> : smaller LLMs produce richer uncertainty signals under C2, yielding tighter TubeNet intervals; and all five judges dramatically outperform the BERT-only baseline, confirming the universal value of LLM uncertainty features.	Confirmed	Gemma2-9b-it (smallest) achieves the narrowest fluency (0.599) and relevance (1.956) intervals, outperforming GPT-OSS-120b on both. Mixtral achieves the narrowest consistency (0.314). Every C2 judge reduces coherence by $\geq 45\%$ and consistency by $\geq 86\%$ vs. Cond. A.	Gemma2 flu: 0.599 vs. GPT-OSS 0.933 (−35.8%). Mixtral cons: 0.314 vs. GPT-OSS 0.325. Worst C2 judge: coh 1.719 (−45%), cons 0.325 (−86%) vs. BERT-only baseline (coh 3.133, cons 2.350).

Table 12: Hypothesis verification (H8–H10): derived from comparison with Sheng et al. (2025) (Table 10) and multi-LLM judge comparison (Table 8). H8 consolidates the width superiority and dimension-specific findings; H9 consolidates coverage reliability and best-method identification; H10 consolidates all judge diversity and universality findings.